FINM3013 Introduction to Financial Derivatives

Chapter 01

Solution to Class exercises

1. B
2. B
3. C. The payoff from the options is 100×(65-60) or $500. The cost of the options is 2×100 or $200. The net profit is therefore 500−200 or $300.
4. A. When the stock price is $35, the two call options provide a payoff of 2×(35−30) or $10. The put option provides no payoff. The total cost of the options is 2×3+ 4 or $10. The stock price in A, $35, is therefore the breakeven stock price above which the position is profitable because it is the price for which the cost of the options equals the payoff.
5. D. When the stock price is $50 the first alternative leads to a position in the stock worth 100×50 or $5000. The second alternative leads to a payoff from the options of 1000×(50−45) or $5000. Both alternatives cost $4000. It follows that the alternatives are equally profitable when the stock price is $50. For stock prices above $50 the option alternative is more profitable.